01 / THE BIG IDEA

Learning from data

An introduction to machine learning

Machine learning allows computers to learn patterns from data, without explicitly programming every rule.

DATA

Examples from the world

MODEL

A pattern we can learn

PREDICTION

An answer for new data

A model learns from examples. Its goal is to generalize.

02 / KEY CONCEPTS

Supervised learning

Learning with a teacher

Training data contains both inputs and known labels.
The model learns the relationship between them.

INPUT	LABEL	TASK
An email message	Spam or not spam	Classify a new email

Classification predicts a category. Regression predicts a number.

03 / MAKING IT WORK

Beyond memorization

Training, testing, and generalization

A model should perform well on examples it has never seen.
Keep a separate test set to measure its real performance.

TRAINING SET

Learn the patterns

TEST SET

Evaluate on unseen data

OVERFITTING

Memorizing the training set

Good performance on training data alone is not enough.